Assignment-9.R

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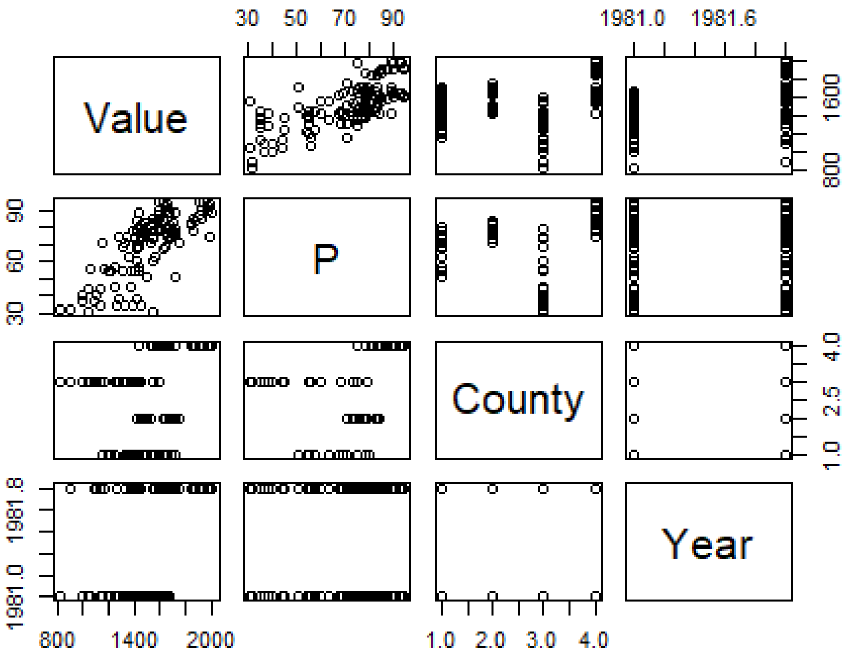
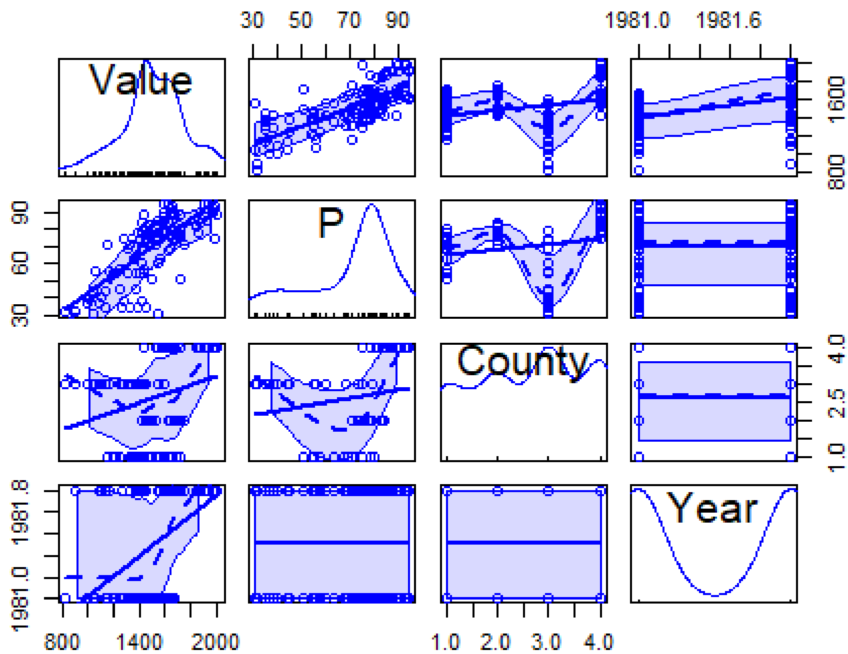
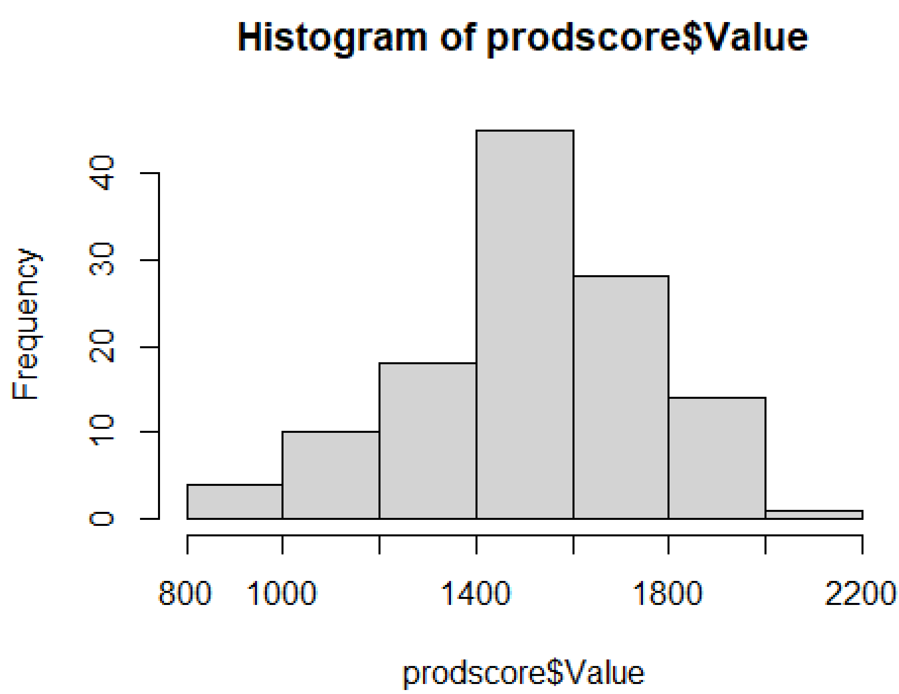
```
### Assignment-9 ####

# 6.19
library(alr4)

data("prodscore")
force(prodscore)

##      County Value  P Year
## 1   LeSueur  1495 51 1981
## 2   LeSueur  1222 54 1981
## 3   LeSueur  1200 55 1981
## 4   LeSueur  1254 57 1981
## 5   LeSueur  1358 63 1981
## 6   LeSueur  1416 67 1981
## 7   LeSueur  1428 69 1981
## 8   LeSueur  1146 71 1981
## 9   LeSueur  1347 75 1981
## 10  LeSueur  1474 75 1981
## 11  LeSueur  1382 78 1981
## 12  LeSueur  1392 78 1981
## 13  LeSueur  1441 80 1981
## 14   Sibley  1652 75 1981
## 15   Sibley  1544 78 1981


# visualizing given data
pairs(~Value+P+County+Year,data=prodscore)
```




```
# From the visualization of the given data I conclude that I see a relation
# between the variables P and the value. we can compare the average assessed
value
# and soil productivity for the given four countries by analyzing the
country,value and P plots.
# The plot year and Value shows that the average assessed value of lands
increased from
# the year 1981-1982.the plot p and year shows soil productivity from 1 to
100(according to the question)
# when comparing the both years the productivity of the soil doesn't change
and I
# see no difference in the projection of observations so it may be a null
plot.

scatterplotMatrix(~Value+P+County+Year,data=prodscore)
```

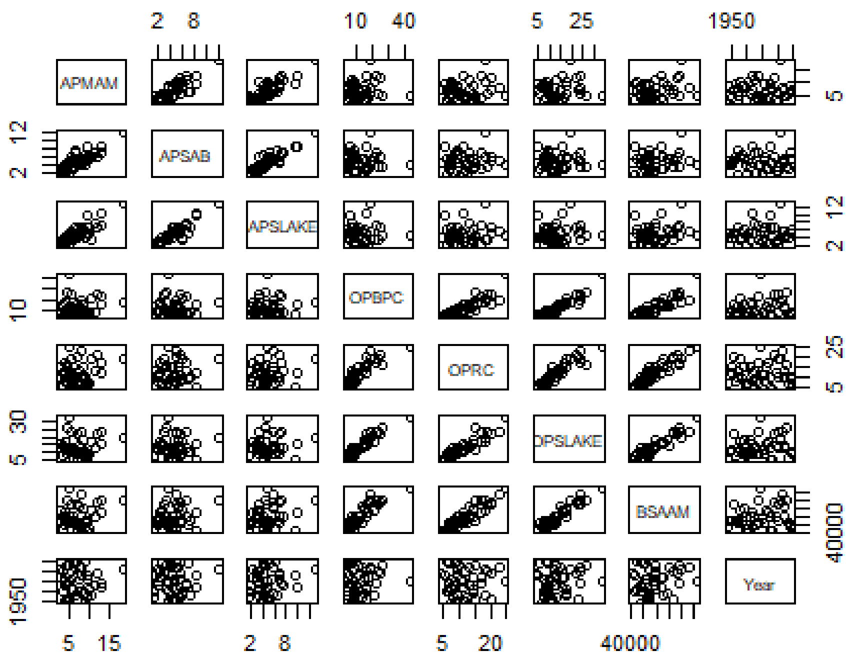



```
hist(prodscore$Value)
```



```r
# our assumptions from the visualization are all correct, but for the plot
# value-p i see a good regression fit with constant variance, the
distribution
# of the data is almost symmetric. so lets dive into the summarizing the
data.

# fitting the regression model
m1 <-lm(Value~P+County+Year,prodscore)
summary(m1)

##
## Call:
## lm(formula = Value ~ P + County + Year, data = prodscore)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -392.77  -53.90   12.16   61.71  265.61
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -4.368e+05  4.043e+04 -10.803  < 2e-16 ***
## P             5.378e+00  1.004e+00   5.356 4.48e-07 ***
## CountyMcLeod  7.164e+01  3.240e+01   2.211    0.029 *
## CountyMeeker -8.534e+01  3.419e+01  -2.496    0.014 *
## CountySibley  1.927e+02  3.550e+01   5.429 3.23e-07 ***
## Year          2.210e+02  2.040e+01  10.830  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 111.7 on 114 degrees of freedom
## Multiple R-squared:  0.8045, Adjusted R-squared:  0.7959
## F-statistic: 93.84 on 5 and 114 DF,  p-value: < 2.2e-16

# According to the summary, the most interested variable "P" has the positive
# effect on Value, if P increases by 1 unit which resulting increase of value
# by 5.378e+00 by taking into account of other regressors.almost all
regressors
# are significant other than some countries.but we want to decide whether the
# variable p is a good regressor or not.
qt(1-0.05/2,114)

## [1] 1.980992

# our t value is 5.356 which is more than the significant level of quantile
α=0.05
# which is 1.980992.The p-value is also approximately zero which means the
variable
# p is significant means it rejects the null hypothesis.so, p has a strong
effect on
# value, it is a good predictor.
# the decision makers can make decisions of taxes on soil productivity land
```

```
by using the the
# average assessed value.


# 8.3
data(water)
# 8.3.1-visualizing the data
pairs(~APMAM+APSAB+APSLAKE+OPBPC+OPRC+OPSLAKE+BSAAM+Year,data = water)
```



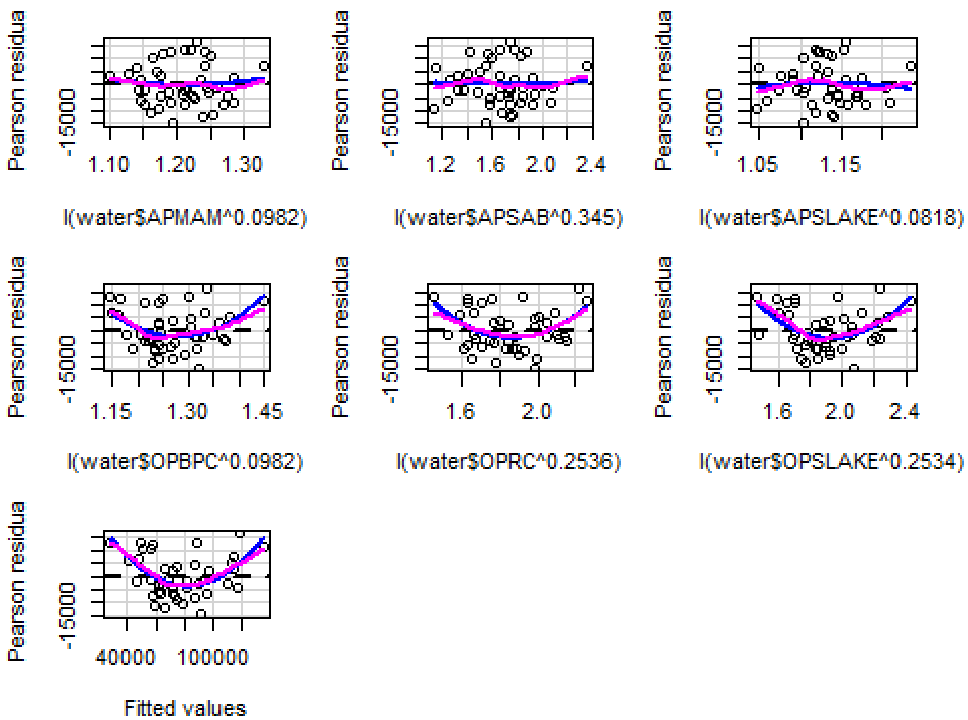

```
# By analyzing the plots, the variables A's have linear models and also have
# constant variances also the same thing with 'O' variables but the variable
BSAAM
# has strong linear relation with 'O' variables. the most common thing is
# all the variable plots are right asymmetric. from variable year i can only
say that
# it has a outlier point but according to the question we should remove it.

# 8.3.2- lets do the power-transformation of box-cox
pt1 <-
powerTransform(cbind(water$APMAM,water$APSAB,water$APSLAKE,water$OPBPC,water$
OPRC,water$OPSLAKE))
summary(pt1)

## bcPower Transformations to Multinormality
##    Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
## Y1    0.0982           0      -0.4625       0.6589
## Y2    0.3450           0      -0.0533       0.7432
```

```
## Y3    0.0818           0      -0.3466       0.5101
## Y4    0.0982           0      -0.2109       0.4073
## Y5    0.2536           0      -0.2255       0.7328
## Y6    0.2534           0      -0.0921       0.5988
##
## Likelihood ratio test that transformation parameters are equal to 0
##  (all log transformations)
##                                        LRT df    pval
## LR test, lambda = (0 0 0 0 0 0) 5.452999  6 0.48716
##
## Likelihood ratio test that no transformations are needed
##                                        LRT df       pval
## LR test, lambda = (1 1 1 1 1 1) 61.20312  6 2.5629e-11
```

```
qt(1-0.05/2,6)
```

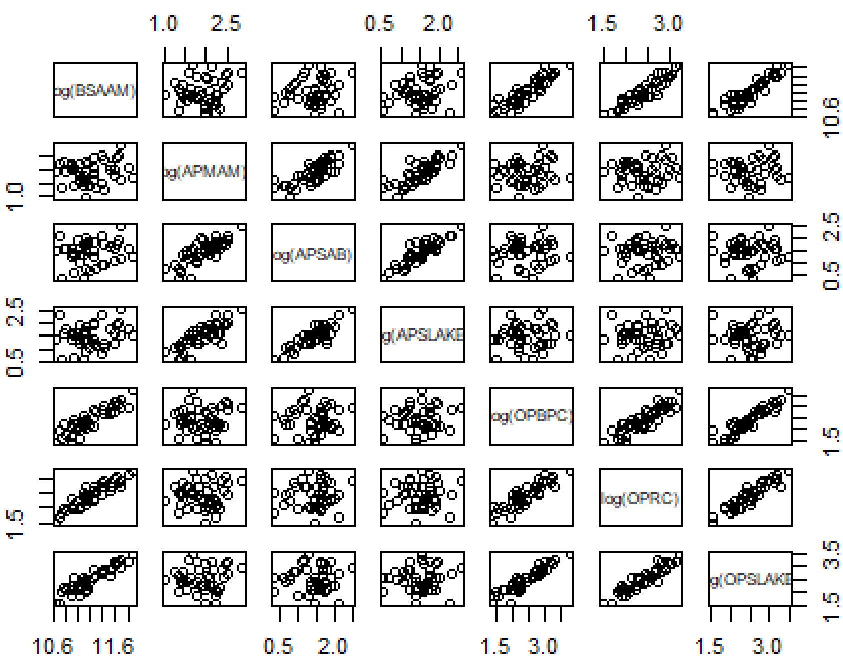
```
## [1] 2.446912
```

```
# the estimated lambda values for the transformed model is given in the
summary
# we also have the confidence intervals of the given all lambdas. according
to
# the hypothesis test(H0=lambda=0) the p value is not significant which means
# we don't reject null hypothesis.so, log model is better which the lamda=0
model
# is better but im not sure because the t value is more than quantilea=0.05
where
# the significant value of quantile is 2.446912.

# lets check the linearity of the predictors using transformed model using
residual plots
m2<-
lm(water$BSAAM~I(water$APMAM^0.0982)+I(water$APSAB^0.3450)+I(water$APSLAKE^0.
0818)+I(water$OPBPC^0.0982)+I(water$OPRC^0.2536)+I(water$OPSLAKE^0.2534))
residualPlots(m2,smooth=TRUE)
```



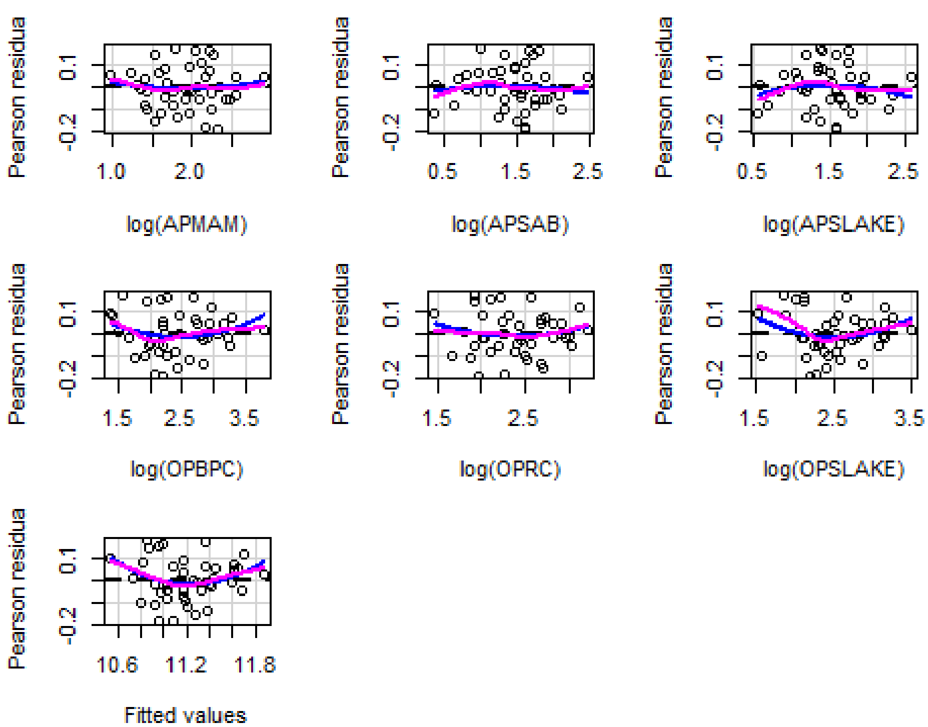

```
##                            Test stat Pr(>|Test stat|)
## I(water$APMAM^0.0982)         0.3995         0.691983
## I(water$APSAB^0.345)          0.1836         0.855418
## I(water$APSLAKE^0.0818)      -0.4351         0.666141
## I(water$OPBPC^0.0982)         2.8196         0.007864 **
## I(water$OPRC^0.2536)          2.8014         0.008234 **
## I(water$OPSLAKE^0.2534)       3.4903         0.001324 **
## Tukey test                    4.2576        2.066e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# by using the function residual plots we can state that for all the A's
predictors
# we get the best linear model which so close to zero which means the
observations are
# symmetric in these models and the transformation works for A's variables.
But for
# the O's and tukey test plots the power transformation doesn't work which
means
# there is quadratic and non parametric effect in the model.so, for these
models
# our plot is not near null.so, its better to find another transformation
models.

# 8.3.3,8.3.4- by analyzing the summary of the power transformation we know
that
# log transformation is better than previous model using hypothesis test. so
```

```
lets
# do the fit the log model and check
m3<-
lm(log(BSAAM)~log(APMAM)+log(APSAB)+log(APSLAKE)+log(OPBPC)+log(OPRC)+log(OPS
LAKE),water)
pairs(log(BSAAM)~log(APMAM)+log(APSAB)+log(APSLAKE)+log(OPBPC)+log(OPRC)+log(
OPSLAKE),water)
```



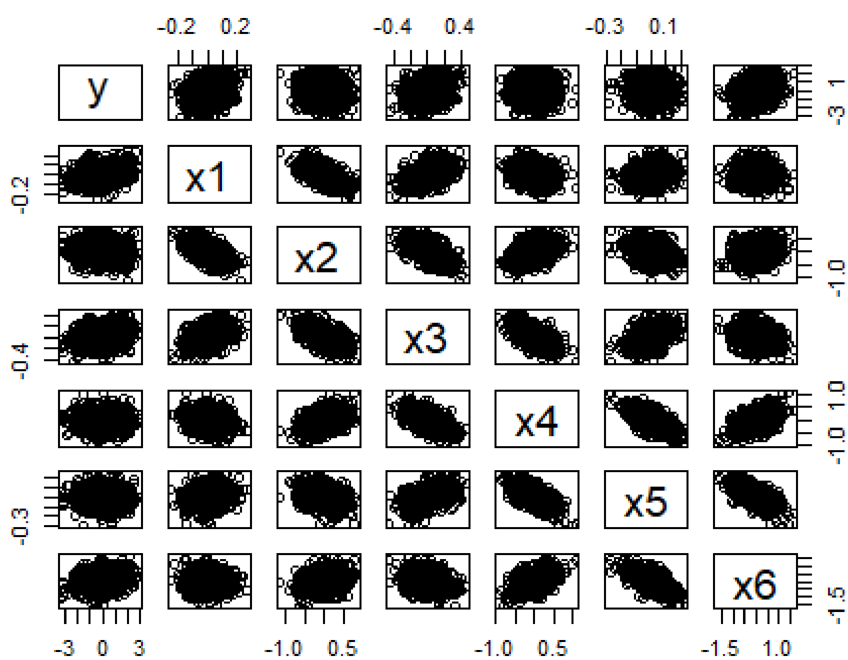

```
# The log transformation for the response log(BSAAM) and all the predictors
because
# i get the strong relation from all the plots associated with the response.
so, the transformation works.

summary(m3)

##
## Call:
## lm(formula = log(BSAAM) ~ log(APMAM) + log(APSAB) + log(APSLAKE) +
##     log(OPBPC) + log(OPRC) + log(OPSLAKE), data = water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18671 -0.05264 -0.00693  0.06130  0.17698
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.46675    0.12354  76.626  < 2e-16 ***
```

```
## log(APMAM)   -0.02033    0.06596  -0.308  0.75975
## log(APSAB)   -0.10303    0.08939  -1.153  0.25667
## log(APSLAKE)  0.22060    0.08955   2.463  0.01868 *
## log(OPBPC)    0.11135    0.08169   1.363  0.18134
## log(OPRC)     0.36165    0.10926   3.310  0.00213 **
## log(OPSLAKE)  0.18613    0.13141   1.416  0.16524
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1017 on 36 degrees of freedom
## Multiple R-squared:  0.9098, Adjusted R-squared:  0.8948
## F-statistic: 60.54 on 6 and 36 DF,  p-value: < 2.2e-16

# in the given summary of the given regression model the negative value coefficients
# are for the log(APMAM)and log(APSAB) and those are -0.02033 and -0.10303. i really
# cannot interpret why i got the negative coefficient only i can say is i see some leverage
# points in the associated plots.so maybe because of these points i get negative values.

# 9.8 the residual function of the given log model
residualPlots(m3,smooth=T)
```



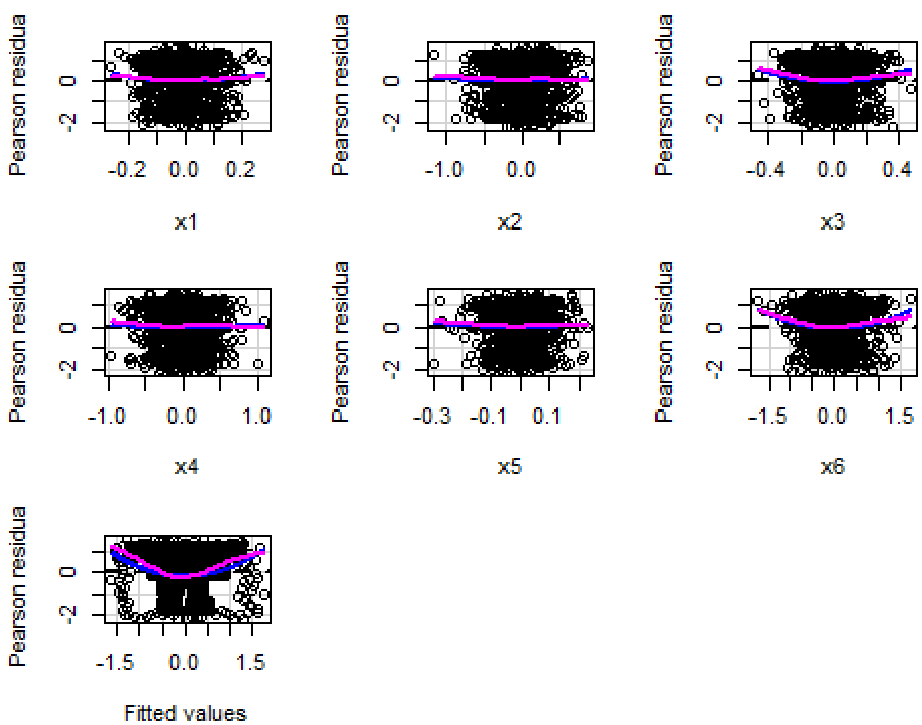

```
##              Test stat Pr(>|Test stat|)
## log(APMAM)      0.4499          0.65553
```

```
## log(APSAB)     -0.4647         0.64502
## log(APSLAKE)   -0.8525         0.39976
## log(OPBPC)      1.3848         0.17487
## log(OPRC)       0.8387         0.40735
## log(OPSLAKE)    1.6295         0.11217
## Tukey test      1.8386         0.06597 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# according the residual plot we get the plots which all almost null plots so
i dont
# see any effect log model for all the variables also for the fitted residual
plot
# but i see the some curvature for the fitted values but most of the valuesd
are
# in the middle so due one or two outliners we get some significance. if we
# remove them then our model is well fitted model which shows no evidence of
the
# quadratic curve.


# 9.1
data(Rpdata)
pairs(y~x1+x2+x3+x4+x5+x6,Rpdata)
```




```
m4<-lm(y~x1+x2+x3+x4+x5+x6,Rpdata)
summary(m4)
```

```
##
## Call:
## lm(formula = y ~ x1 + x2 + x3 + x4 + x5 + x6, data = Rpdata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1977 -0.7631  0.1729  0.8851  1.6359
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.02481    0.03188   0.778    0.437
## x1           4.14061    0.50954   8.126 1.32e-15 ***
## x2           1.01233    0.15522   6.522 1.11e-10 ***
## x3           3.99614    0.32663  12.234  < 2e-16 ***
## x4           0.96045    0.16657   5.766 1.09e-08 ***
## x5           3.75122    0.64726   5.796 9.17e-09 ***
## x6           0.95390    0.08561  11.142  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.003 on 983 degrees of freedom
## Multiple R-squared:  0.3112, Adjusted R-squared:  0.307
## F-statistic: 74.03 on 6 and 983 DF,  p-value: < 2.2e-16

residualPlots(m4,smooth=T)
```



```
##            Test stat Pr(>|Test stat|)
## x1            1.4098         0.158930
## x2            0.4535         0.650309
## x3            2.5134         0.012117 *
## x4            0.2375         0.812337
## x5            0.6086         0.542951
## x6            3.2239         0.001306 **
## Tukey test    7.4126        1.238e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```